MIS 430 – 580N – Emerging Issues – Cyber Security – Spring 2026. Instructor: Thi Tran.

Group Project 1 (Possible points: 100)

Topic: Data Visualization for Cyber Security

Group Name: 2

Session (MIS 430.01 – 580N.01): 430.01

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Instructions:

Refer back to all of the slides and explanations in Week 4, Week 6 and Week 7 to do this project.

Submit this assignment on BrightSpace.

Due date: Midnight Sunday, March 22, 2026.

Only submit Word document (.doc or .docx) or .pdf file.

Only one member of each group must submit the file.

Save the file with your name at the beginning, such as

Group [X] _ Group Project 1 _ MIS 430 / MIS 580N – Spring 2026.docx

Instructions:

1. **Find a dataset with more than 200 lines of records related to any of the Cyber Security concerns / issues, such as: cyberattacks, fake news, cyber intrusions, identity thief, malware, etc.**
2. **Perform basic descriptive analytics, including but not limited to: average/mean, median, standard deviation of differences across records, outlier detections, distributions of the majority of the records. You can show the numerical results here then visualize in the graphs later.**
3. **Reason for the inclusion or exclusion of or special focuses on outliers: why the outliers are noises (not important but distracting values) or the main focuses (such as detecting abnormal signals, like frauds or hackers' abnormal behaviors).**
4. **Ideally: You should run the following tasks with and without the outliers (full versus filtered datasets) and compare and contrast the results. If you only consider 1 (full or filtered version), clearly reason for the choice.**

5. **Choose a tool for visualization, or a few tools. Make sure to specify in your submission which tool(s) you use. For example: Tableau, R-Studio, python, etc.**
6. **Make <u>at least 5 different graphs / visualizations</u> that involve from 1 to many variables. For example: histogram, scatterplot, pie chart, boxplot, line chart, bar chart, or stacked bar chart. Specify each of your charts with the type and a short description, such as: "Pie chart of the groups of victims of…".**
7. **If you use programming languages, include the source codes at the end of the deliverable.**
8. **Explain the key findings after visualizing for <u>each</u> of the charts / graphs. Including: (1) what are interesting findings; (2) how such findings can help people better understand the situations; (3) how such findings can suggest practical decisions or strategies.**
9. **<u>ONLY FOR MIS 580N:</u> Write from 1 to 3 paragraphs to summarize the interesting / key / important findings from <u>all</u> previous data visualization tasks. Connect the findings to the current, emerging technical situations related to the chosen context, such as the use of Artificial Intelligence (AI), Internet of Things (IoT), automatic data analysis, and more. Aim to reason for: (1) how the findings about Cybersecurity can strengthen the use of some related technologies; (2) how the technologies can help resolve the Cybersecurity issues that you detected or concern about.**

<u>Grading rubrics:</u>

- **Directly and sufficiently answer all questions: up to 80%.**
- **Proven extra efforts in critical reasonings, justifications and arguments: up to 20%.**
- **Be specific when answering the applications of concepts and knowledge, including but not limited to examples, detailed applications, use cases, etc.: up to 50%.**
- **Professional references (clearly listed in "References" section at the end of the deliverable) in APA format with all details (authors, titles, venues, years): up to 15%.**
- **Professional in-text citations associated with specific points in deliverable with proper APA format, including author names and year of publication: up to 15%.**
- **Authentic and verifiable references and citations: violation (including incorrect references in ANY details mentioned above) will face from 50% deduction to serious academic dishonesty cases.**
- **Unmatched references and citations but not falsified details: up to 20% deduction.**
- **Failure to cite LLMs used, if applicable: up to 20% deduction.**

<u>Beginning your work from here, retaining previous instructions:</u>

Cybersecurity has become an increasingly critical concern as digital systems continue to expand and cyber threats grow in sizes and complexity. Phishing attacks, in particular, exploit subtle patterns in URLs to deceive users and gain unauthorized access to sensitive information. As a result analyzing URL characteristics through data-driven approaches is essential for detecting and preventing such threats.

This project examines a cybersecurity dataset containing 662,592 observations, analyzing phishing behavior through five key perspectives. First, the relationship between HTTP usage in phishing is explored to evaluate whether secure protocols truly indicate safety. Second, numeric density and URL length are analyzed to determine how quantitative characteristics influence phishing likelihood. Third, statistical testing is applied to assess the strength and significance of relationships between URL length and phishing classification. Fourth, structural elements of URLs including number of dots and dashes are examined to understand how attackers manipulate link complexity. Finally, the presence of special symbols, such as the "@" character, is analyzed alongside path length to identify subtle but high-risk indicators.

By combining descriptive analytics, statistical methods, and multiple visualization techniques, this project aims to uncover patterns that distinguish phishing URLs from legitimate ones. The findings provide a comprehensive view of how different URL features interact and offer practical insights for improving phishing detection and cybersecurity decision-making.

#1 Data Visualization of Phishing Links (HTTP Vs. HTTPS)

Overview of Variables

This section examines the relationship between the HttpsInHostname variable and the Phishing variable in the dataset. The HttpsInHostname variable shows whether the term "https" appears in the host name of a URL, while the Phishing variable shows whether the website is classified as phishing or legitimate. These two variables are useful to compare because they help show both how common HTTPS-related wording is in the dataset and how phishing is distributed across HTTPS and non-HTTPS links. For this part of the project, the data visualizations were created in Python, including a heatmap and bar-chart comparisons, to make the patterns in the data easier to understand.

Descriptive Statistics

The dataset contains 629,845 observations. Out of those, only 9,946 websites (1.58%) contain HTTPS in the host name, while 619,899 websites (98.42%) do not. This shows that HTTPS in the host name is very uncommon in this dataset. The phishing variable is also unevenly distributed, although not as extremely. 99,856 websites (15.85%) are classified as phishing, while 529,989 websites (84.15%) are classified as non-phishing. These descriptive statistics are important because they provide context for the rest of the analysis and show that the dataset is especially imbalanced when it comes to the HTTPS variable.

Table 1: Summary of Descriptive Statistics for Binary Variables

Variable	Mean	Median	Standard Deviation	Interpretation

HTTPS in Host Name (0 = No, 1 = Yes)	0.0158	0	0.125	Only 1.58% of websites contain HTTPS in the host name, showing a highly imbalanced distribution.
Phishing (0 = No, 1 = Yes)	0.1585	0	0.365	About 15.85% of websites are phishing, showing a less extreme but still uneven distribution.

Since both variables are binary, the descriptive statistics need to be interpreted a little differently than they would be for continuous data. The mean of the HTTPS variable is 0.0158, which simply means that 1.58% of the URLs contain HTTPS in the host name. The mean of the Phishing variable is 0.1585, showing that phishing websites make up 15.85% of the dataset. In both variables, the median is 0, which tells us that the typical observation falls in the "No" category. The standard deviation is 0.125 for HTTPS and 0.365 for Phishing, showing that phishing status varies more across the dataset, while the HTTPS variable is concentrated heavily near zero because it appears so rarely. In this case, the most important issue is not a traditional numerical outlier, but the strong imbalance in how often HTTPS appears.

Category Distribution and Cross-Tabulation

Because the HTTPS category is so small, proportions are more useful than raw totals when comparing phishing behavior. Among the 9,946 websites that contain HTTPS in the host name, 7,621 are phishing and 2,325 are not phishing. On the other hand, among the 619,899 websites that do not contain HTTPS in the host name, 92,235 are phishing and 527,664 are not phishing. This means that about 76.6% of the URLs in the HTTPS category are phishing, compared to about 14.9% of the non-HTTPS group. Looking at the proportion within each category gives a much more accurate picture than looking only at total counts.

Table 2: Cross-Tabulation of HTTPS in Host Name and Phishing Status

HTTPS in Host Name	Is Phishing	Is Not Phishing	Total
Yes	7,621	2,325	9,946
No	92,235	527,664	619,899
Total	99,856	529,989	629,845

Table 3: Phishing Rate Within Each HTTPS Category

HTTPS in Host Name	Phishing Rate
Yes	76.6%
No	14.9%

These percentages show why the proportional comparison matters. If the analysis only focused on raw totals, the non-HTTPS group would dominate the results simply because it contains almost all of the records in the dataset.

Visualization Interpretation

The heatmap helps make this pattern more visible. Even though websites with HTTPS in the host name make up only a very small portion of the dataset, the heatmap shows that phishing cases are heavily concentrated within that category compared to its size. A bar chart comparing phishing rates by category supports the same conclusion by showing that the phishing rate is much higher in the HTTPS group than in the non-HTTPS group. At the same time, a bar chart based on total counts helps explain the overall size difference between the two groups. Together, these visualizations make it easier to understand both the imbalance in the dataset and the way phishing is distributed across the two categories.

Interpretation of the Results

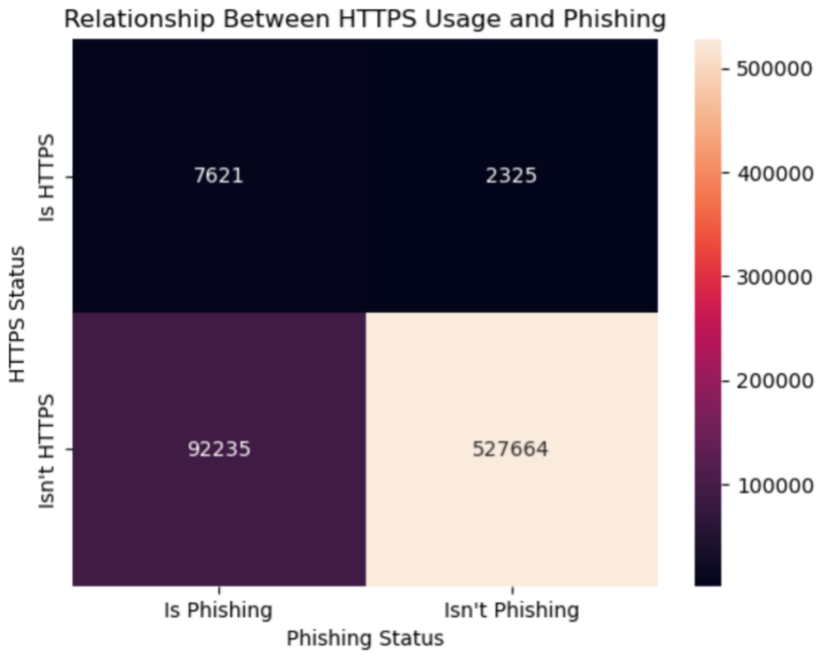
This pattern may appear counterintuitive, as HTTPS is commonly associated with secure and trustworthy websites. However, existing research shows that phishing attackers increasingly adopt HTTPS to make malicious websites appear more credible and effective (Kim et al., 2021). Additionally, industry reports indicate that a large majority of phishing sites now use SSL/TLS encryption, demonstrating that attackers actively exploit HTTPS as a trust signal rather than a barrier (Anti-Phishing Working Group [APWG], 2020). The widespread availability of free and automated certificate services has significantly lowered the barrier for implementing HTTPS, enabling attackers to mimic legitimate websites more easily (Aas, 2025).

Cybersecurity Implications

Overall, these findings have important implications for cybersecurity awareness and decision-making. People often assume that HTTPS automatically means a site is safe, but the results here show that this is not always true. In this dataset, HTTPS in the host name appears rarely, but when it does appear, it is associated with a much higher phishing rate. This suggests that attackers may use HTTPS-related wording to make websites seem more trustworthy at first glance. Because of that, users and organizations should not rely on HTTPS alone when deciding

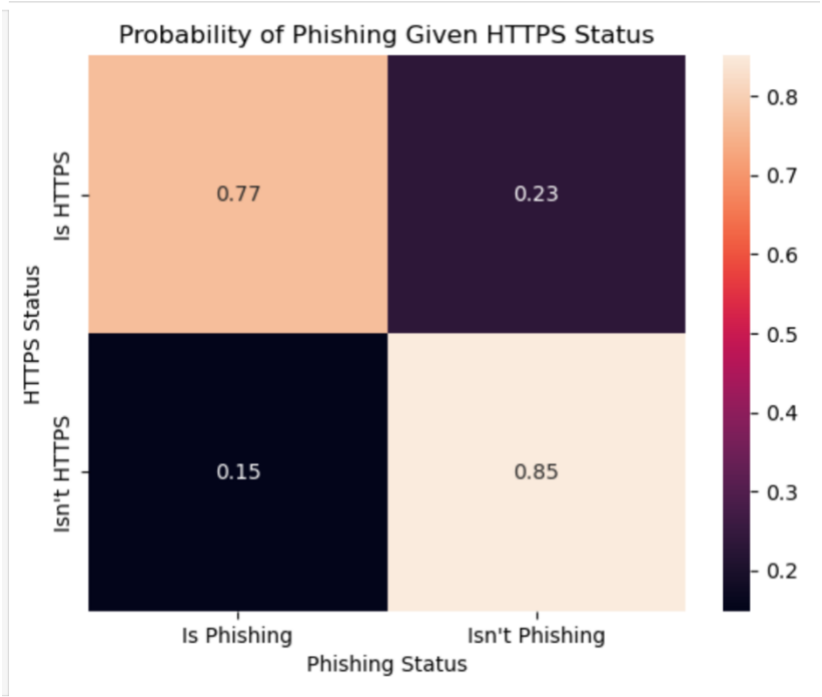
whether a website is legitimate. Instead, they should also pay attention to other warning signs such as suspicious domains, misleading page content, and unusual website behavior.

Data Visualizations



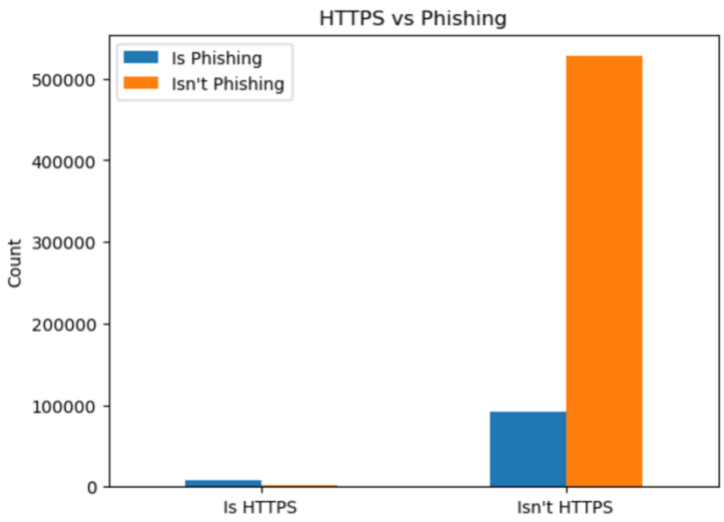
Visualization 1 (Heatmap: HTTPS vs Phishing)

This heatmap visualizes the distribution of phishing and non-phishing websites across HTTPS and non-HTTPS categories. An interesting finding is that although HTTPS-enabled websites make up a very small portion of the dataset, they contain a disproportionately high number of phishing instances. This helps us better understand that phishing is not evenly distributed and is more concentrated within HTTPS websites in this dataset. The visualization is important because it makes this imbalance clearly visible, which would be difficult to detect from raw numbers alone. From a practical standpoint, this suggests that users and organizations should not assume that HTTPS guarantees safety and should instead rely on additional indicators when evaluating website legitimacy.



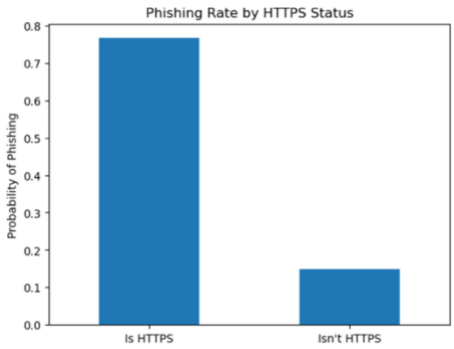
Visualization 2 (Proportion/Rate Comparison – Phishing Rate by HTTPS)

This visualization compares the proportion of phishing websites within HTTPS and non-HTTPS categories rather than total counts. A key finding is that the phishing rate is significantly higher among HTTPS-enabled websites (approximately 76.6%) compared to non-HTTPS websites (approximately 14.9%). This helps us better understand the relative risk associated with each category by removing the distortion caused by the dataset's imbalance. By focusing on proportions, this visualization provides a more accurate representation of phishing likelihood within each group. From a practical perspective, this suggests that cybersecurity decisions and user awareness training should emphasize that HTTPS alone is not a reliable indicator of safety and that additional verification methods are necessary.



Visualization 3 (Bar Chart: Phishing Counts by HTTPS Status)

This bar chart compares the total number of phishing and non-phishing websites across HTTPS and non-HTTPS groups. An important finding is that the majority of websites in the dataset do not use HTTPS, and most non-phishing instances fall within this category due to its large size. This helps us understand the overall distribution of the dataset and highlights the significant imbalance between HTTPS and non-HTTPS usage. The visualization is useful because it provides context for interpreting other graphs, particularly those that focus on proportions rather than raw counts. From a practical standpoint, this suggests that analysts must account for data imbalance when evaluating cybersecurity risks to avoid misleading conclusions based solely on total counts.



Visualization 4 (Phishing Counts by Protocol)

This bar chart focuses specifically on phishing websites and compares the number of phishing instances between HTTPS and non-HTTPS categories. A key finding is that most phishing websites are associated with non-HTTPS simply because that category dominates the dataset; however, HTTPS websites still represent a meaningful portion of phishing cases despite their low overall frequency. This helps us better understand that phishing occurs across both

protocol types and is not limited to insecure websites. The importance of this visualization is that it isolates phishing behavior, making it easier to analyze how attacks are distributed. From a practical perspective, this suggests that cybersecurity strategies should address threats across both HTTPS and non-HTTPS environments and should not rely solely on protocol type as an indicator of risk.

#2 Analysis of Numeric Density in Phishing URLs

The dataset displays phishing website detection features, where each column shows characteristics of a website URL. Phishing is when websites pretend to be legitimate. It takes someone's personal information by acting like a real website, and then it attacks by stealing money or the identity of somebody. The important variables that will be analyzed and later visualized are "NumNumericChar", which is the number of numeric characters in the URL. Then, "URL Length", that gives the total number of characters in the URL. Finally, "Phishing", identified as 1 = phishing and 0 = legitimate. These features help in the arrangement patterns in URLs because it shows how phishing attackers often use excessive numeric characters to copy real domains or conceal harmful elements. Comparing these variables depicts the analysis of numeric density in URLs when it comes to phishing. Ultimately, looking at the number of numbers packed into a URL relative to its length. In order to see if the composition of numbers impacts the probability of phishing to happen more often in a website.

Descriptive Analysis:

Tool - Excel

Variable	Mean	Median	Std Dev
NumNumericChar	0.65	1.00	1.43
URL Length	60.25	48.00	44.60

Figure 1. An Excel table that shows the results I got from calculating the variables of "NumNumericChar" and "URL Length" mean, median, and standard deviation.

To begin, "NumNumericChar" distribution is slightly left-skewed since the median(1) is greater than the mean (0.65). The calculation found is surprising and is very interesting to see because it is counter-intuative because usually in phishing it is expected for the mean to be greater than the median. In this instance, it doesn't follow the normal route and is non-obvious. This explains that most URLs contain a 0 or 1 numeric character, but since there are multiple 0s it brings down the mean. Typically, in phishing there are two types of strategies. The first is low numeric usage in the website URL. An example of this can be venmo-login.com. While the second type is high numeric usage in a URL, such as venmo-login-pay3426789.com. The dataset proves that phishing doesn't always need high numeric usage, which isn't your typical finding when looking at phishing examples. For the standard deviation for "NumNumericChar" it is 1.43 which is high compared to the mean of 0.65. This can be interpreted that the data is widely spread and some URLs have more numbers than the typical.

Moving on, to the findings of the "URL Length" variable, which is the opposite of "NumNumericChar" because this data is what is typically seen/expected in phishing datasets. The mean (60.25) is greater than the median (48), so it is right skewed. Phishing attackers would hide malicious parts deep in the URL, extending its length in like the parameters section of URL. The standard deviation in this case is very high with the amount being 44.60. To clarify this means that the URLs range from short (most likely clean) to extremely long (more suspicious). Many of the URLs contain very few numbers, while some have extremely large numeric counts. These extreme values are potential indicators of phishing behavior. All in all, the main takeaway is that numeric characters aren't always shown, but when there is a high amount that is a strong phishing signal.

Outlier Detection:

Tool - Excel

NumNumericChar				Key	
Calcaulte Quartiles	0	1	-->	Q1 = QUARTILE (range,1)	Q3 = QUARTILE (range,3)
IQR	1		-->	IQR = Q3 - Q1	
Outlier Threshold	-1.5	2.5	-->	LB = Q1 - 1.5*IQR	UB = Q3 + 1.5*IQR

URL Length				Key	
Calcaulte Quartiles	32	77	-->	Q1 = QUARTILE (range,1)	Q3 = QUARTILE (range,3)
IQR	45		-->	IQR = Q3 - Q1	
Outlier Threshold	-35.5	144.5	-->	LB = Q1 - 1.5*IQR	UB = Q3 + 1.5*IQR

Figure 2. Calculations of outliers by using IQR in variables "NumNumericChar" and "URL Length"

Detecting outliers within columns "NumNumericChar" and "URL Length" in the dataset was done by using IQR (Interquartile Range) in Excel. By following the formulas in the key, it calculates the quartiles, IQR, and outlier threshold of upper/lower bound. The data shows that in the first quartile(Q1) also known as the 25th percentile of data for "NumNumericChar" = 1, and "URL Length" = 32, reveals the median of the lower half of the dataset. For the third quartile(Q3) which is the 75th percentile, the values above give the median of the upper half of the dataset. For the IQR formula, it helps calculate the spread of the middle 50% of data. In the instances above, the higher IQR means the middle 50% of data is more spread out. The last calculation was to determine the lower/upper outlier bounds. Any number that is outside of these bounds mostly means it is a potential outlier in the dataset.

Investigating deeper, the outliers in "NumNumericChar" that are greater than 2.5 numeric characters represent bot-generated links, automated phishing URLs, and obscure attack patterns. For "URL Length" the outliers that are greater than 144.5, are very suspicious and lead to phishing attempts. When running analysis with outliers the results prove there is a higher mean in URL Length, larger variance, and stronger separation between phishing and legitimate URLs. The outlier data between the variables proves that URL length is a stronger/more consistent phishing indicator than numeric character alone. Although numeric characters by

itself aren't as consistent in presenting phishing URLs, the combination of longer URL structures significantly increases the likelihood of malicious intent.

Overall, outliers are important signs to show abnormal attacker behavior, so outliers should be included in the main analysis. Running analysis without outliers demonstrates smaller standard deviation, a narrower distribution, and less visible difference among phishing and regular URLs. To conclude, removing outliers limits the visibility of harmful behavior, so using the full dataset is ideal.

Visualization:

Tool - Tableau

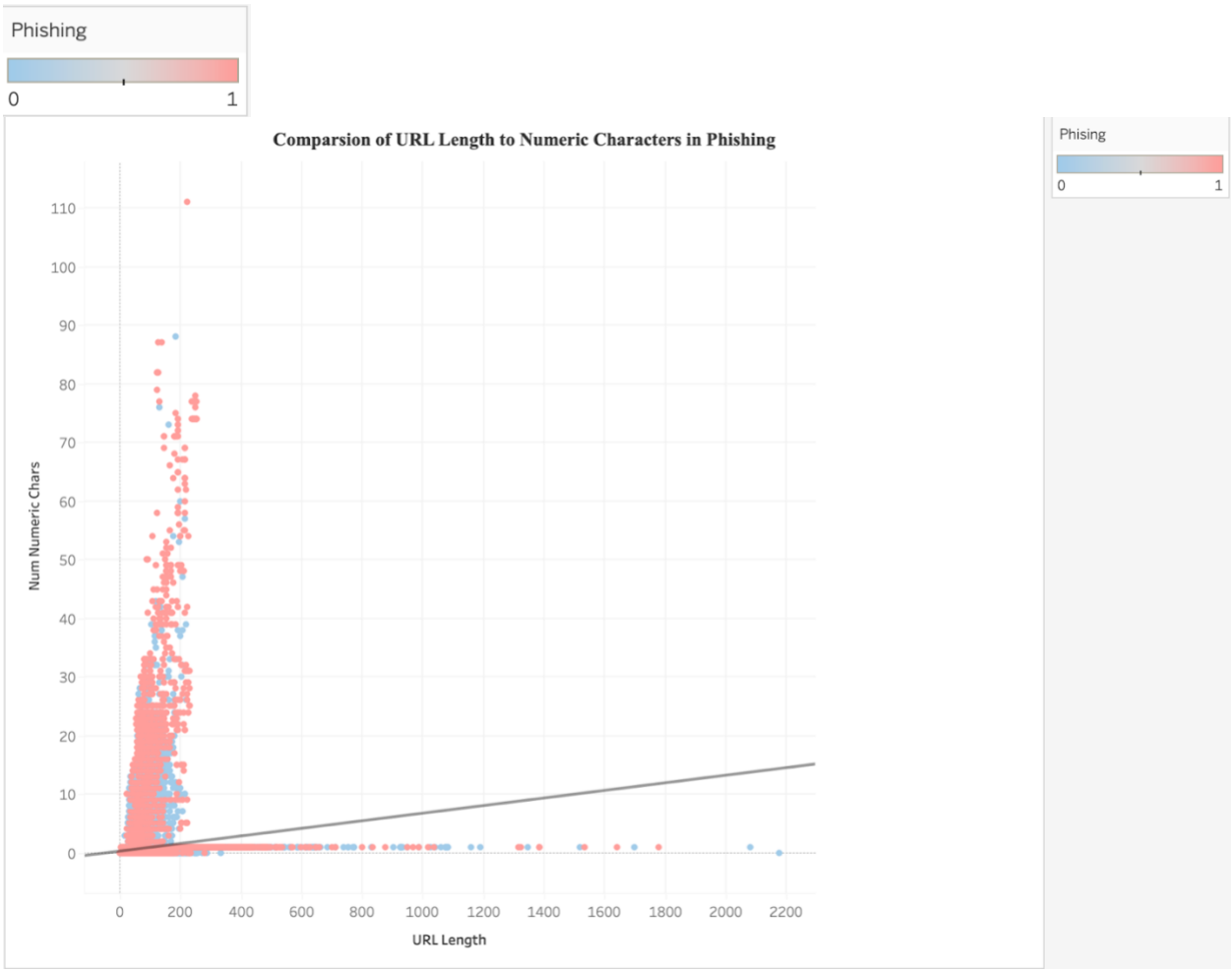


Figure 3. Scatter Plot comparing URL Length and NumNumericChar to Phishing

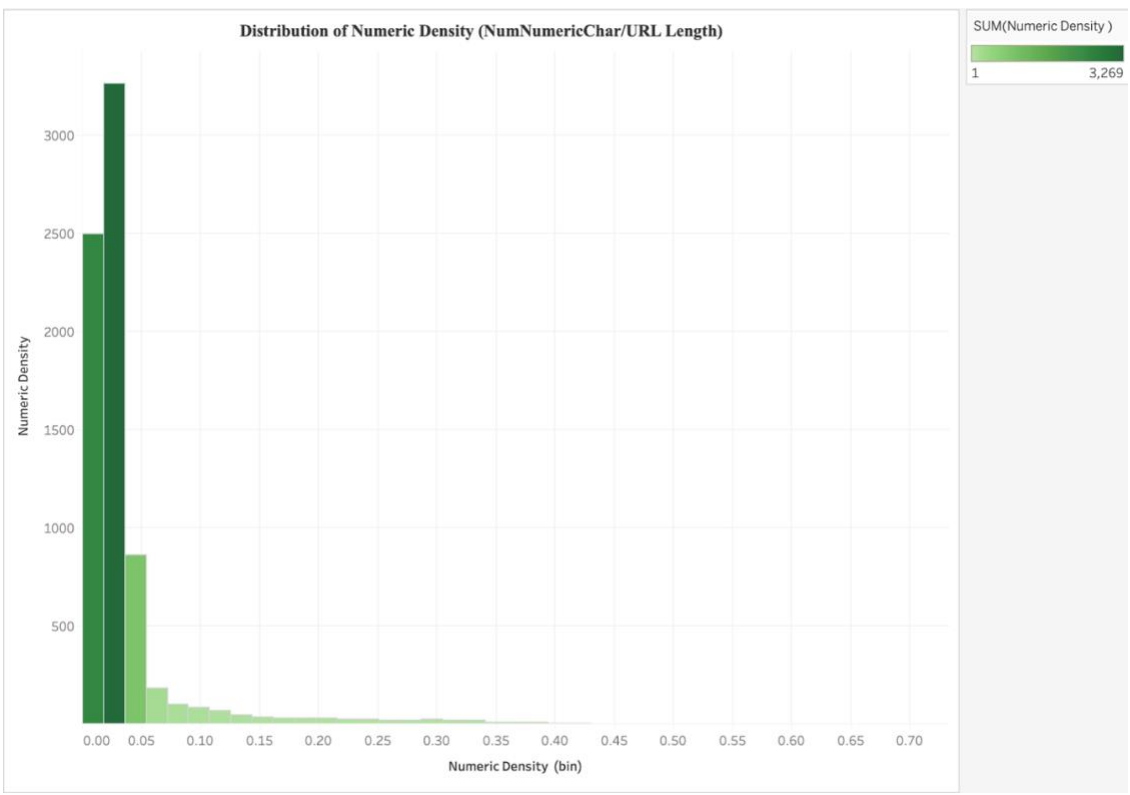
Initially, at first glance of the scatter plot, visually it shows the points being clustered at the bottom left corner. Translating to a high number of low numeric characters with middle to low URL lengths. As far as outliers go, there are a few and they represent high numeric characters and high URL lengths, which constitute more with phishing. In the legend the red dots are phishing = 1, while the blue dots are legitimate = 0. The graph's axis range was cut

short, in order to focus on where the bulk of data lies. Exploring further, there is a trend line that stays relatively low in the graph. In this line it contains this data:

Num Numeric Chars = 0.00647589*Url Length + 0.262342
R-Squared: 0.0407973
P-value: < 0.0001

R-Squared is a form of regression analysis. It measures the strength of a relationship also known as the goodness of fit. It ranges between 0 to 1, and in this case, it is 0.0407973. Since it is closer to 0 there is a lack of consistency. For the P-value it gives the probability of obtaining test results. The data captures it to be <0.0001, which makes it a low P-value since it is lower than 0.05. This suggests the predictor is important and the relationship is statistically significant.

Additionally, the highest URL length shown in the range of the graph is 2,175 and that correlates with no phishing and is legitimate. The highest value of numeric characters in this range is 111 and correlates with phishing. In the scatterplot it proves that there is always a greater URL length compared to numeric characters which is expected, since URL length makes up all the components of a URL including numbers. Although something that is interesting is if URL tends to have more numeric components it correlates greater with phishing, and the same thing applies to the length. However, the graph shows that there can still be phishing and no phishing in low numeric regions. This relates back to the different strategies of complexity that are applied by hackers.



SUM(Numeric Density)
1
3,269

Figure 4. - Histogram of Numeric Density Distribution (NumNumericChar / URL Length)

Next up, the histogram prominently shows a higher concentration of numeric density in the earlier ranges of data. This is an important graph to interpret because it was made by creating a field in tableau by dividing "NumNumericChar" by "URL Length" to get the numeric density between the two variables. The y axis range was shortened to capture a closer look at the data area that is more filled. The legend displays the darker the green as the most common, and the lighter green as the least common to be assorted in groups in the selected data.

Looking at the x axis which is labeled as Numeric Density(bin) is created from numeric measures. The bin represents a range of values on the x-axis and the count (frequency) that is displayed on the y-axis. In data transformation, bins convert a continuous measure into a discrete dimension, enabling you to count how many data points fall within each specific range. In the graph above the greatest density falls into the 0.02 bin range and 3,269 data values are shown in the numeric density. The higher the density the more suspicious and inclined to phishing a URL is. This is beneficial to know for those working in companies to teach employees so no data can be harmed, by clicking on a harmful website/email.

Final Takeaways:

To wrap up, the findings of the dataset prove 4 main things to be true. The first is that URL length is a strong phishing indicator, because the greater the length is the more likely it leads to this attack. This is expected and makes sense for attackers to incorporate longer URLs by encrypting more to the length. The second finding is that analyzing numeric characters alone isn't reliable because there is a mix of phishing and legitimate URLs with number composition. Even though when more numbers are displayed it does correlate greater with phishing. The third discovery is that the most powerful signal to determine phishing is combining numeric characters with URL length to see phishing results. These two variables work well together and detect phishing activity greatly. The last bit that I was able to analyze is that outliers in the dataset aren't a bad thing but are helpful and represent the real attacks that aren't expected, so using it when determining descriptive analytics was important.

Regarding practical applications, in cybersecurity systems rules to follow such as URL Length being greater than 100 mark it to be flagged since it is very suspicious behavior. As well as numeric density being high, make sure that is flagged. This project is a big lesson learnt for companies to pay attention to, since analyzing what has happened in this dataset can apply to various businesses. Companies should implement greater filters like checking for spam and phishing detection when it comes to websites and emails that they received. Since an employee may fall in a trap with linked information by clicking on faulty websites or emails.

Now connecting phishing back to relevant real-world examples, according to "Phishing or Not Phishing?" article, it discusses various companies doing surveys that prove phishing is a massive threat. Pulling stats such as phishing attacks being the most common cyber incident to European small to mid-sized businesses. Or that 86% of organizations had at least one user connect to a phishing site. These high probabilities of phishing attacks mean it shouldn't be undermined and must be made aware for all users. Stakeholders should be mindful before investing into a business that the firm is following safety procedures so that data can't be taken and used for attacks. Stakeholders can also influence their expertise and put their control in

fixing safety procedures that aren't as focused on like phishing to cybersecurity software. As mentioned in, “Tracking phishing attacks over time”, many attacks stay active throughout the observation period, so avoiding these attacks in the first place is great to protect a business. Knowing information that correlates with phishing like high URL lengths/numeric characters in a link can help this recognition. Lastly, for any users online avoid long URLs and number-infested links.

#3 Comparing the relationship with URL length and phishing instances.

One of the most surprising outcomes of this analysis was how consistently the data challenged common assumptions about what a phishing URL looks like. Most people would assume that phishing links are long, cluttered, and visually suspicious. However, the data showed the opposite. Phishing URLs averaged 46.9 characters compared to 63.0 for legitimate URLs, and the Very Short bin accounted for the largest absolute volume of phishing URLs at 74,070 cases. At the same time, the Pearson correlation of -0.1298 confirms that URL length alone is a weak predictor, and the U-shaped risk pattern visible in the heatmap shows that both extremely short and extremely long URLs carry elevated phishing rates. This suggests that attackers do not follow a single approach. Some keep URLs short and simple to avoid raising red flags, while others use long, complex strings to hide the true destination. A detection system that relies on URL length as a single feature would miss both strategies. The bigger takeaway is that phishing detection needs to account for multiple URL characteristics at once, including length, path depth, query parameters, and domain structure, rather than treating any one feature as a definitive signal on its own.

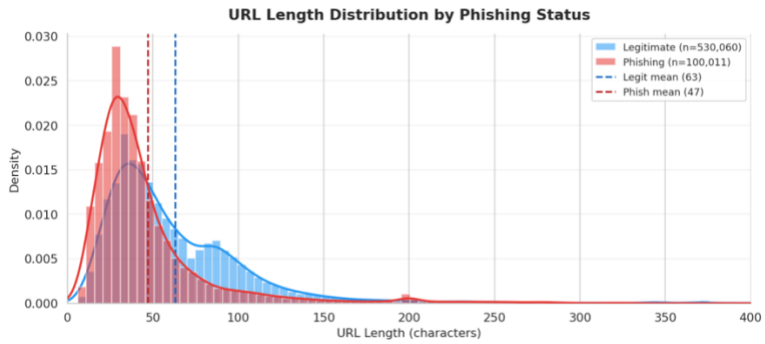
```
=== Phishing Rate by URL Length Bin ===
```

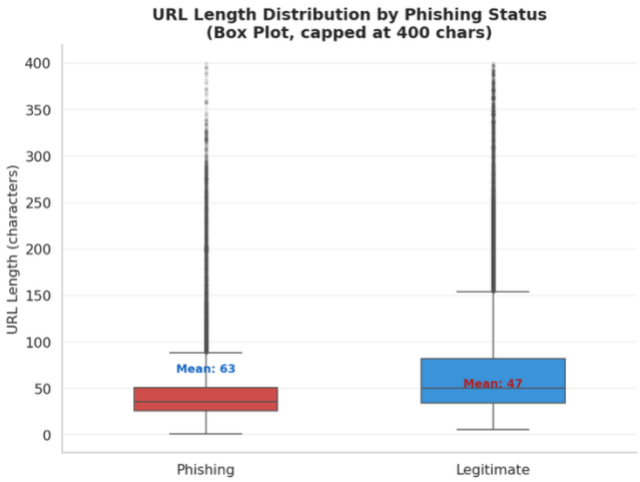
LengthBin	Total	Phishing	Legitimate	Phishing_Rate_%
Very Short (≤50)	339236	74070	265166	21.83
Short (51–100)	210380	18882	191498	8.98
Medium (101–200)	69991	5356	64635	7.65
Long (201–500)	10331	1669	8662	16.16
Very Long (500+)	133	34	99	25.56

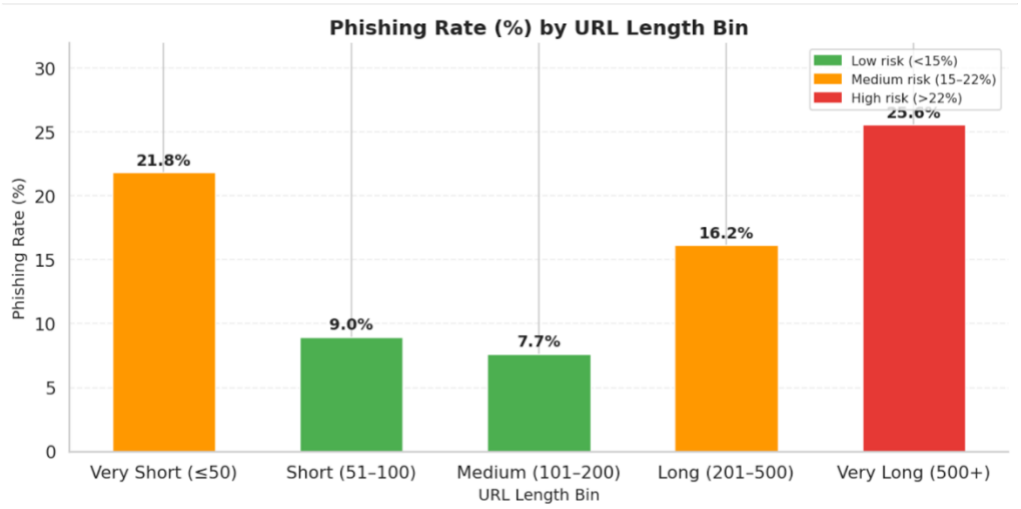
```
=== URL Length Summary Statistics ===
```

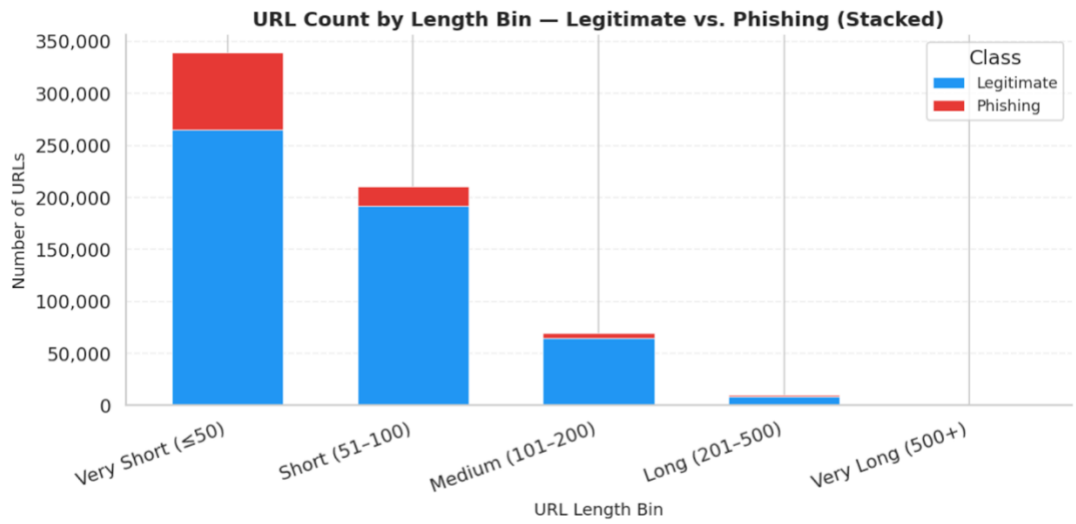
Label	count	mean	std	min	25%	50%	75%	max
Legitimate	530060.0	63.00	45.15	6.0	34.0	50.0	82.0	2175.0
Phishing	100011.0	46.92	43.48	1.0	26.0	36.0	52.0	1779.0

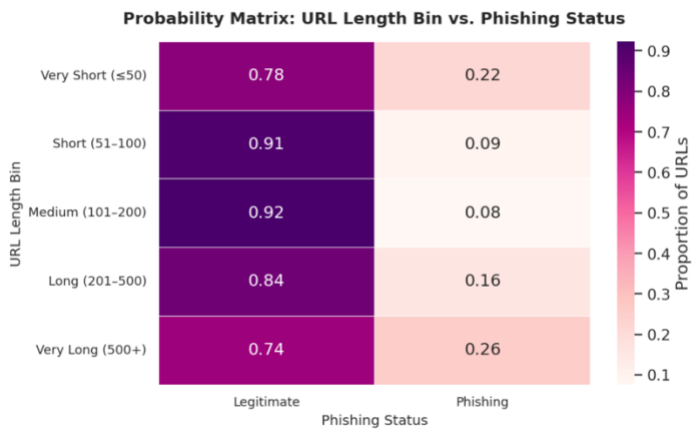
```
Mean URL Length  -->  Legitimate: 63.0 chars  |  Phishing: 46.9 chars
Median URL Length -->  Legitimate: 50 chars  |  Phishing: 36 chars
Pearson Correlation (UrlLength <--> Phising): -0.1298
```


URL Length Distribution by Phishing Status
Legitimate (n=530,060)
Phishing (n=100,011)
Legit mean (63)
Phish mean (47)
Density
URL Length (characters)


URL Length Distribution by Phishing Status
(Box Plot, capped at 400 chars)
URL Length (characters)
Mean: 63
Mean: 47
Phishing
Legitimate


Phishing Rate (%) by URL Length Bin
Low risk (<15%)
Medium risk (15–22%)
High risk (>22%)
Phishing Rate (%)
21.8%
9.0%
7.7%
16.2%
25.6%
Very Short (≤50)
Short (51–100)
Medium (101–200)
Long (201–500)
Very Long (500+)
URL Length Bin






```
============================================================
        URL LENGTH vs. PHISHING — KEY FINDINGS
============================================================
  Total URLs analyzed     : 630,071
  Legitimate              : 530,060 (84.1%)
  Phishing                : 100,011 (15.9%)

  Avg length — Legitimate : 63.0 chars
  Avg length — Phishing   : 46.9 chars
  Difference              : 16.1 chars (legit URLs are LONGER)

  Pearson correlation     : -0.1298 (negative = longer -> less phishing)
  Highest-risk bin        : Very Long (500+) -> 25.6% phishing rate

  INTERPRETATION:
  Contrary to common assumptions, phishing URLs in this
  dataset tend to be SHORTER than legitimate ones. Very
  short URLs (≤50 chars) have a ~22% phishing rate, and
  very long URLs (500+) spike again at ~26%. The safest
  range is 51–200 characters (~8–9% phishing rate).
============================================================
```

Statistical Analysis of URL Length and Phishing

To move beyond descriptive observations, several formal statistical tests were conducted to evaluate the relationship between URL length and phishing status across the 630,071 URLs in this dataset.

A Welch's t-test was used to compare the mean URL length between legitimate and phishing URLs, accounting for the unequal variances between the two groups. The results were statistically significant ($t = 106.58$, $p < 0.001$), confirming that the difference in average URL length between legitimate URLs (mean = 63.0 characters) and phishing URLs (mean = 46.9 characters) is not due to chance. To assess the practical significance of this difference, Cohen's d was calculated at 0.36, which falls in the small-to-medium range. This means that while the difference is real and statistically reliable, URL length alone explains only a modest portion of the variation between classes, which is an important distinction to make given the large sample size.

Since URL length is not normally distributed in either class, a non-parametric Mann-Whitney U test was also conducted as a robustness check. The result ($U = 34{,}978{,}147{,}304$, $p < 0.001$) confirmed that the two distributions are significantly different from one another, reinforcing the t-test findings without relying on normality assumptions.

A point-biserial correlation was computed between URL length and the binary phishing label, yielding $r = -0.1298$ ($p < 0.001$). The negative direction confirms that longer URLs are associated with a lower probability of being phishing in this dataset. However, the magnitude of the correlation is weak, which is consistent with the Cohen's d result. Statistical significance here is driven largely by the sample size of over 600,000 records, so the effect size is the more meaningful figure for practical interpretation.

To examine whether phishing rates differ significantly across URL length categories, a chi-square test of independence was performed on a 5x2 contingency table comparing URL length bins against phishing status. The result was highly significant ($X2(4) = 20{,}076.22$, $p < 0.001$), confirming that phishing rates are not evenly distributed across length categories. Cramer's V, which measures the strength of association, was calculated at 0.18, indicating a weak but consistent relationship. The conditional phishing rates per bin tell the more interesting story: URLs in the Short (8.98%) and Medium (7.65%) bins sit well below the overall dataset phishing rate of 15.87%, while the Very Short (21.83%) and Very Long (25.56%) bins exceed it considerably.

Across all four tests, the relationship between URL length and phishing status is statistically significant but modest in practical effect size. The data does not support using URL length as a standalone classifier. What it does support is using URL length as one feature within a broader detection framework, particularly flagging URLs below 50 characters and above 500 characters as warranting additional scrutiny. The convergence of parametric and non-parametric results adds confidence that these findings are robust and not artifacts of distributional assumptions.

#4 Analysis of URL Structural Characteristics: Number of Dots and Dashes against Phishing

Descriptive Analytics of URL structure

Phishing websites frequently manipulate the structure of URLs to make suspicious links appear legitimate. This analysis examines two structural characteristics in the dataset: number of dots and number of dashes in the URLs provided in the dataset. Many dots can indicate the use of subdomains, designed to mask the true domain name, while multiple dashes are often used to mimic brand names or create misleading addresses.

Descriptive analytics of the number of dots and dashes were first calculated to understand the distribution of these variables across the dataset. The dataset consists of 662,592 values. All descriptive analytics were performed in Excel.

Descriptive Analysis of NumDots

	Mean	Median	Minimum	Maximum	Range	Standard Deviation	Missing Values
NumDots	2.19	2	0	42	42	1.48	0

The descriptive statistics above for NumDots indicate that the mean value is 2.19, with a median of 2. The similarity between mean and median values indicates that most records in the NumDots column contain a relatively small and consistent number of dots. However, the presence of extremely high values indicates that some URLs contain an unusually large number of dots. The minimum value is 0, while the maximum value is 42, resulting in a range of 42. The standard deviation of 1.48 suggests that there is a moderate variability in the number of dots across different URLs.

Outlier analysis using the Interquartile Range (IQR) method through Excel revealed a noticeable number of extreme values in the NumDots columns. Specifically, approximately 11,084 observations were identified as outliers, representing 1.67% proportion of the whole dataset records. This suggests that while most URLs contain a typical number of dots, a subset of values includes a significantly higher number of dots. In the context of phishing detection, this pattern is important to keep in the analysis. Phishing URLs are often intentionally structured with an excessive number of symbols to appear legitimate or to hide the true domain, which naturally results in higher dot count. Therefore, these outliers are important to keep in the analysis, since they likely reflect genuine phishing strategies rather than data inaccuracies.

Descriptive Analysis of NumDashes

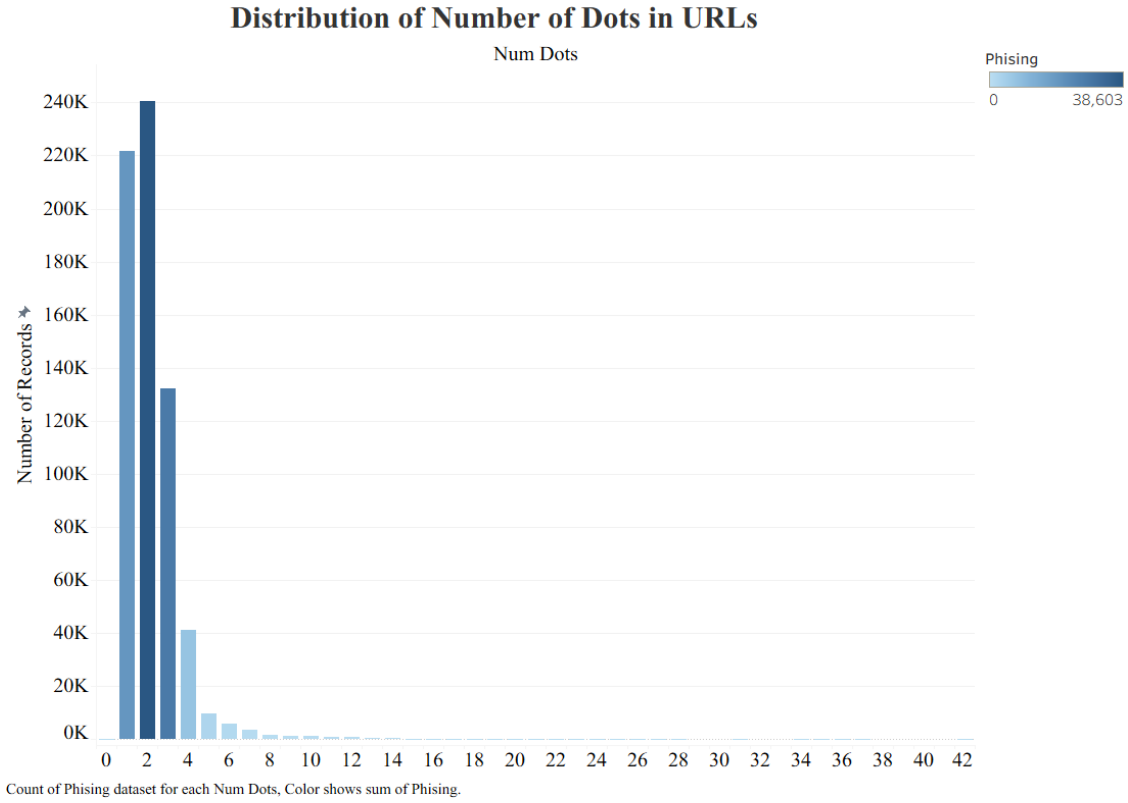
	Mean	Median	Minimum	Maximum	Range	Standard Deviation	Missing Values
NumDashes	1.55	0	0	87	87	2.97	0

On the other hand, the descriptive statistics for NumDashes indicate that the mean value is 1.55, while the median is 0. The median of 0 indicates that more than half of the URLs in the dataset contain no dashes. The minimum value is 0 and the maximum value is 87, resulting in a range of 87. The standard deviation of 2.97 suggests that there is a wider dispersion in the number of dashes compared to the number of dots. The big difference between mean and median values indicates a skewed distribution, where most URLs contain few or no dashes, but a small number of URLs contain extremely large values.

Similarly, the NumDash variable showcases a larger number of outliers, with around 61,635 records. This proportion of 9.3% indicates a likely skewed distribution, where most URLs contain a few dashes, but a considerable number include unusually large numbers. It is important to keep outliers in the analysis, because this pattern is consistent in phishing techniques, where cyberattackers use multiple dashes to create long and complex URLs that mimic trusted sources. Removing these outliers would eliminate critical patterns associated with phishing behavior and reduce the dataset's ability to capture extreme, but important cases.

Visualization Analysis

Graph 1: Histogram of Distribution of Number of Dots in URLs



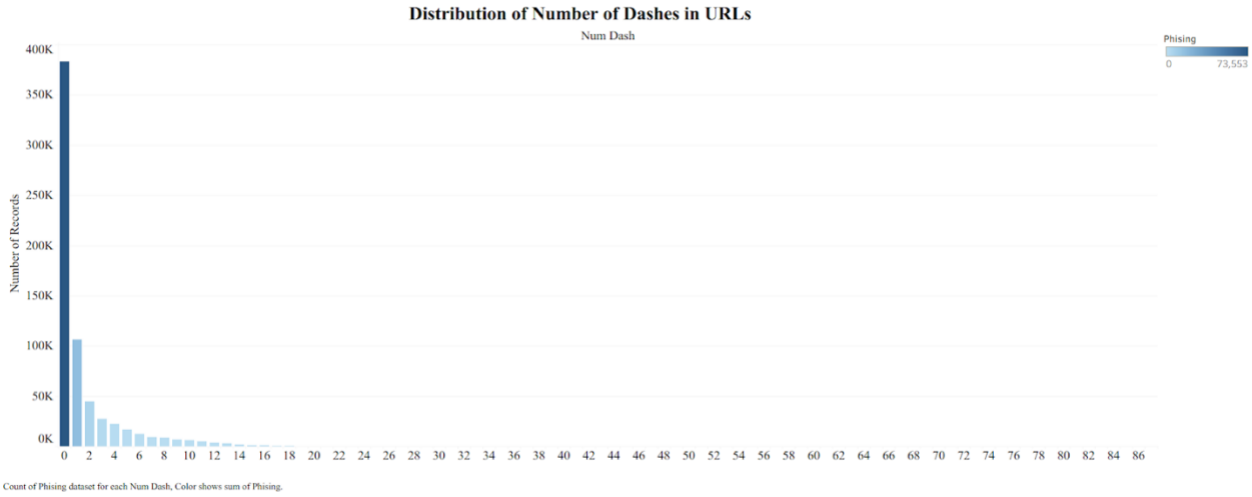
The histogram below illustrates the distribution of the number of dots across all URLs in the dataset. Most of the URLs contain between 1 and 3 dots, with the highest concentration around 2

dots. This finding directly aligns with the calculated mean of 2.19 and median of 2. Additionally, this indicates that most URLs in the dataset follow a relatively simple domain structure.

The graph is right-skewed, extending toward higher values. A smaller number of the URLs contain an unusually high number of dots, reaching up to the maximum of 42. These extreme values represent outliers, which as mentioned above, are important to keep in the analysis to reflect the phishing purpose. Additionally, the color intensity represents phishing frequency and suggests that phishing URLs are more present among higher dot counts.

The key findings from the first graph are that while most URLs are simple, unusually high numbers of dots are associated with suspicious behavior. The visualization helps to improve the understanding of how phishing URLs are structured and suggests that cybersecurity systems can flag URLs with excessive dots as potentially harmful, improving detection accuracy.

Graph 2: Histogram of Distribution of Number of Dashes in the URLs



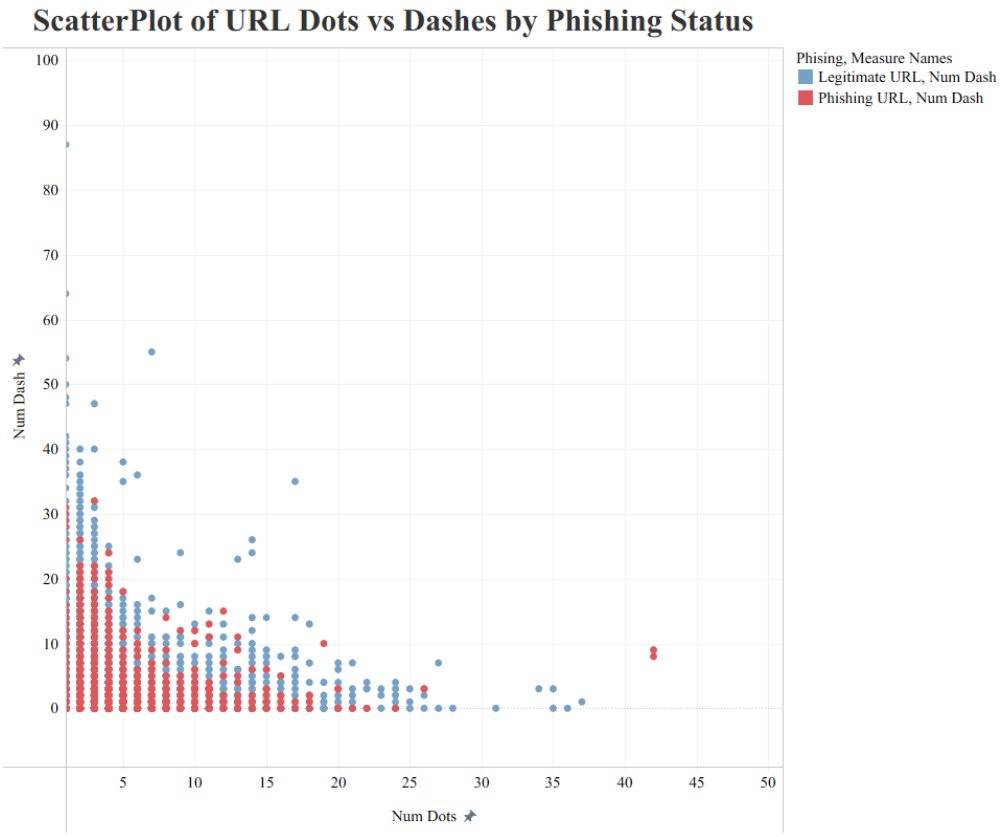
The histogram illustrates the distribution of the number of dashes in URLs. Most URLs in the dataset contain zero dashes, which is consistent with the median value of 1. This indicates that most legitimate URLs tend to avoid dashes and maintain simpler structures.

At the same time, the distribution is right-skewed, similarly to the number of dots graph. The long tail extending to the maximum of 87, indicating the presence of extreme outliers. Compared to the dots, the spread of dashes is much wider, which aligns with the higher standard deviation 2.97.

The color intensity reveals that phishing URLs are more concentrated with a higher number of dashes. This supports the idea of keeping the outliers in the analysis, and that attackers use dashes to imitate brand names or create complex and misleading links.

The key finding is that excessive use of dashes is a strong indicator of phishing behavior. It helps distinguish between regular and harmful URL structure and suggests that detection systems can assign higher risk scores to URLs containing multiple dashes.

Graph 3: Scatter Plot of Number of Dots vs Number of Dashes by Phishing Status



Legitimate Dots and Phishing Dots vs. Num Dash. Color shows details about Phising and Num Dash. The view is filtered on Phising, which excludes Null.

The scatter plot below illustrates the relationship between the number of dots and dashes, differentiated by phishing (red) and legitimate (blue) URLs. Most observations are concentrated in the lower-left corner of the graph, where both the number of dots and the number of dashes are relatively low, indicating that simple URL structures are most common.

As the number of dots and dashes increases, the data points become more dispersed. Phishing URLs appear more frequently in areas with higher numbers of dashes and moderate to high dot counts, suggesting that more complex URL structures are associated with phishing behavior. Additionally, extreme values such as identified outliers are more visibly associated

with phishing observations, reinforcing the idea of unusually complex URLs being strong indicators of phishing attacks. However there is still some overlap between phishing and legitimate URLs.

The key findings from the graph demonstrate that increased complexity in the structure of the URL is directly linked to a higher likelihood of phishing. The graph provides a deeper understanding of how multiple URL features ,such as the quantity of dots and dashes, interact and suggests that combining variables can improve phishing detection.

Conclusion

Phishing detection research has shown that URL-based features play critical roles in recognizing harmful websites from legitimate ones. Lexical characteristics of URLs, such as their structure and composition, are commonly used in machine learning models to improve detection accuracy (Alazaidah et al., 2026). According to Kritika (2025) legitimate URLs typically contain a smaller number of dots, usually around two, while phishing URLs often include additional subdomains to block the true domain, resulting in a higher number of dots. Furthermore, attackers frequently use dashes in domain names to add misleading prefixes that imitate legitimate brand names and deceive users. Patterns like this highlight how increased URL complexity is associated with phishing attempts (Kritika, 2025).

The findings above are consistent with the results of this analysis. The dataset showed that most URLs contained a low number of dots and dashes, while the outliers are associated with increased phishing practice. Therefore, variables such as the number of dots and dashes are not only statistically significant, but also practically useful for identifying phishing attempts and improving cybersecurity decision systems.

#5 Analysis of @Symbol and Path Length:

Descriptive Statistics

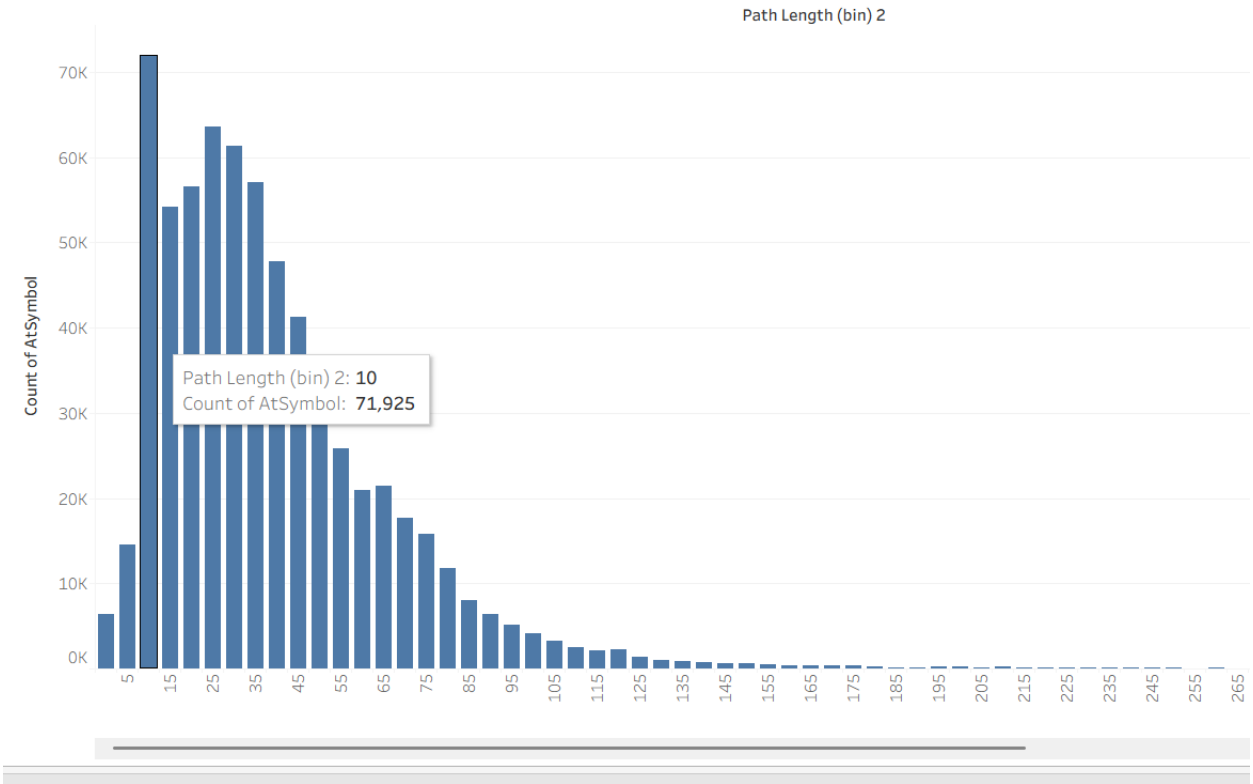
Variable	Mean	Median	Standard Deviation
AtSymbol	0.00219	0.00000	0.05376
PathLength	40.99606	35.00000	29.72243

The AtSymbol values show that the @ symbol is extremely uncommon across the dataset. A mean of 0.00219 indicates that this feature is present in only a small percentage of URLs, while a median of 0 confirms that it is not present in the typical URL. The variable is useful in phishing analysis due to its unusual behavior, which acts as a warning sign rather than an ordinary URL characteristic. Moreover, the standard deviation of AtSymbol is 0.05376. Given that this field is binary, that value indicates a narrow spread around zero, implying that the vast majority of observations do not contain the symbol. In descriptive terms, this supports the conclusion that the @symbol is uncommon, but when it appears, it is significant. The variable should be interpreted as a unique and significant suspicious feature, rather than something that changes over time across URLs.

The PathLength results offer a different type of insight. The mean path length is 40.99606, with a median of 35. The mean is greater than the median, so the distribution is right-skewed. This means that most URLs cluster in the short-to-moderate range, but a small number of much longer paths push the average higher. The PDF histogram shows the tallest bars in the lower and middle bins, gradually tapering off as path length increases. The standard deviation for PathLength is 29.72243, which is large in comparison to the mean. Path lengths vary significantly across the dataset, rather than being tightly grouped around a central value. In practical terms, phishing-related URLs do not all have the same structure. Many are moderate in length, while others are significantly longer and more layered. This variability highlights the importance of not relying solely on a single path-length threshold to detect phishing.

Taken together, these descriptive statistics support a more careful interpretation of the relationship between the two variables. The @ symbol is uncommon in general, so when it appears, it serves as a stronger warning. In contrast, path length varies greatly and should be interpreted as a contextual feature. Longer paths do not necessarily lead to more phishing. However, suspicious symbols like @ become especially important in URLs with moderate structural complexity.

Visual 1: Histogram (Tableau)



The Tableau histogram enhances the descriptive analysis of PathLength by displaying the numerical summary in visual form. The descriptive statistics indicated a mean path length of 40.996, a median of 35, and a standard deviation of 29.722. Together, these values indicate that the distribution is not perfectly balanced. Because the mean is greater than the median, the data is right-skewed, which means that while many URLs have short or moderate path lengths, a small number of much longer URLs push the average up. The histogram visually confirms the pattern. The majority of observations are concentrated in the lower and middle path-length bins, with the highest frequency around 10-20, and the counts steadily decrease as path length increases. The long tail on the right side of the histogram matches the idea that unusually long paths are present, but they are much less common than shorter ones.

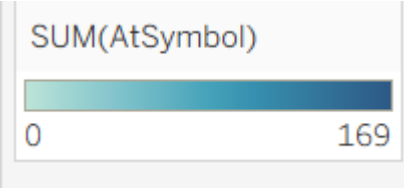
When this is linked to AtSymbol, the interpretation becomes even more significant. The descriptive statistics for AtSymbol revealed a mean of 0.00219, a median of 0, and a standard deviation of 0.05376. This indicates that the @ symbol is extremely rare in the dataset, appearing in only approximately 0.219% of observations. Since the median is 0, most URLs lack this feature entirely. Even though it is uncommon, the @ symbol remains significant because it is a known suspicious feature in phishing URLs. When combined with the histogram of path length,

the analysis suggests that phishing indicators are more than just extreme path size. Instead, suspicious URLs may still appear in the more common short-to-moderate path-length ranges, making them appear more plausible to users. This supports the interpretation that phishing URLs do not need to be extremely long to be deceptive; instead, suspicious elements may be embedded within links that still appear structurally normal to users (Herzberg & Jbara, 2008). In this sense, the histogram shows that many URLs fall into normal-looking structural ranges, whereas the AtSymbol statistic demonstrates that rare but suspicious characters can still distinguish risky URLs within those ranges.

Overall, the visual and descriptive statistics combine to demonstrate that PathLength describes the overall spread and structure of URLs, whereas AtSymbol highlights a rare but potentially misleading feature. The histogram describes the shape of the PathLength data, the mean and median explain its skew, and the standard deviation demonstrates its variability. Meanwhile, AtSymbol statistics show that this feature is uncommon, making its appearance stand out when it does occur. This combined interpretation is useful for Project 1 because it remains descriptive while also explaining what the data patterns indicate about phishing-related URL design.

Visual 2: Heatmap (Tableau)

Path Len..	
0	39
5	25
10	48
15	67
20	105
25	91
30	83
35	96
40	169
45	72
50	83
55	108
60	99
65	75
70	35
75	38
80	25
85	27
90	13
95	4
100	15
105	7
110	5
115	10
120	20
125	11
130	9
135	3
140	19
145	0
150	1
155	10
160	3



The heat map shows how AtSymbol occurrences are distributed across PathLength values. The darkest cell appears at a path length of around 40, where the sum of AtSymbol reaches 169, indicating that the presence of the @ symbol is more concentrated in specific middle path-length ranges rather than being evenly distributed across all URLs. This supplemented the descriptive statistics. AtSymbol's mean of 0.00219, median of 0, and standard deviation of 0.05376 indicate that the feature is extremely rare and absent from the vast majority of observations. However, the heat map demonstrates that when the @ symbol appears, it is more likely to occur within specific path-length values rather than randomly throughout the dataset. For PathLength, the mean of 40.996 being greater than the median of 35 indicates a right-skewed distribution, while the standard deviation of 29.722 shows considerable variability. The concentration of AtSymbol counts near moderate path lengths indicates that suspicious URLs do not have to be extremely long to appear deceptive. Herzberg and Jbara (2008) reinforce this interpretation by showing that users often rely on superficial signs of legitimacy, which makes moderate-looking but structurally suspicious URLs more difficult to identify at first glance. Instead, phishing-related indicators may appear in URLs with similar-looking structures, making them more difficult to identify at first glance.

The analysis concludes that PathLength and AtSymbol represent distinct but complementary aspects of URL design related to phishing. While the @ symbol serves as an uncommon but possibly more potent indicator of manipulation, PathLength captures structural variation. Herzberg and Jbara (2008) demonstrate that users often do not identify misleading URL features, particularly when deceptive elements are incorporated into links that retain a credible overall appearance. The @ symbol's presence within moderate path lengths is particularly significant in this context because it implies that phishing URLs may be made to look structurally normal while still containing subtle signs of deception. When combined, these results demonstrate that structural features and uncommon symbolic markers should be taken into account when detecting suspicious URLs.

Conclusion

In conclusion, this project demonstrates that phishing detection is a complex problem that cannot be addressed through a single variable or assumption. Across all five analyses, different URL characteristics revealed distinct but complementary insights into phishing behavior. The HTTPS analysis showed that secure protocols do not guarantee safety, as attackers increasingly exploit HTTPS to build trust. The numeric density and URL length analysis highlighted that while individual features may be inconsistent predictors, their combination provides stronger signals of malicious intent.

Statistical testing further confirmed that although relationships such as URL length and phishing are statistically significant, their practical impact is limited when considered in isolation. Structural analysis of dots and dashes demonstrated that increased URL complexity is strongly associated with phishing attempts, while the examination of special symbols like the "@" character revealed that rare but abnormal features can serve as powerful warning signs even within otherwise normal-looking URLs.

Overall, the findings emphasize that effective phishing detection requires a multi-step approach that integrates structural, numerical, and symbolic indicators. Outliers across all analyses were shown to represent meaningful attack patterns rather than noise, reinforcing their importance in cybersecurity modeling. From a practical standpoint, organizations should implement detection systems that evaluate multiple URL features simultaneously and train users to recognize both obvious and subtle signs of phishing. As cyber threats continue to evolve, leveraging comprehensive data analysis and visualization techniques will remain essential for improving security and reducing vulnerability to attacks.

Appendix A: #1 Data Visualization of Phishing Links (HTTP Vs. HTTPS) Python Code used

In [23]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import chi2_contingency
```

In [3]:
```
df = pd.read_csv("Phising_dataset_predict.csv")

print(df.head())      # Shows first 5 rows
print(df.info())      # Shows column names and data types
print(df.shape)       # Shows (rows, columns)
```

```
   Unnamed: 0  NumDots  UrlLength  AtSymbol  NumDash  NumPercent  \
0           0        3         72         0        0           0
1           1        3        144         0        0           0
2           2        3         58         0        0           0
3           3        3         79         0        1           0
4           4        3         46         0        0           0

   NumQueryComponents  IpAddress  HttpsInHostname  PathLevel  PathLength  \
0                   0          0                0          5          44
1                   2          0                0          3          16
2                   0          0                0          2          24
3                   0          0                0          6          50
4                   0          0                0          4          29

   NumNumericChars  Phising
0                0      1.0
1               41      1.0
2                0      1.0
3                0      1.0
4                2      1.0
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 662591 entries, 0 to 662590
Data columns (total 13 columns):
 #   Column              Non-Null Count   Dtype
---  ------              --------------   -----
 0   Unnamed: 0          662591 non-null  int64
 1   NumDots             662591 non-null  int64
 2   UrlLength           662591 non-null  int64
 3   AtSymbol            662591 non-null  int64
 4   NumDash             662591 non-null  int64
 5   NumPercent          662591 non-null  int64
 6   NumQueryComponents  662591 non-null  int64
 7   IpAddress           662591 non-null  int64
 8   HttpsInHostname     662591 non-null  int64
 9   PathLevel           662591 non-null  int64
 10  PathLength          662591 non-null  int64
 11  NumNumericChars     662591 non-null  int64
 12  Phising             630071 non-null  float64
dtypes: float64(1), int64(12)
memory usage: 65.7 MB
None
(662591, 13)
```

In [10]:
```
Not_Secure_Not_Phishing = ((df["HttpsInHostname"] == 0) & (df["Phising"] == 0)).sum
Not_Secure_Phishing     = ((df["HttpsInHostname"] == 0) & (df["Phising"] == 1)).sum
Secure_Not_Phishing     = ((df["HttpsInHostname"] == 1) & (df["Phising"] == 0)).sum
Secure_Phishing         = ((df["HttpsInHostname"] == 1) & (df["Phising"] == 1)).sum

print("HTTPS=0 & Phishing=0:", Not_Secure_Not_Phishing)
print("HTTPS=0 & Phishing=1:", Not_Secure_Phishing)
print("HTTPS=1 & Phishing=0:", Secure_Not_Phishing)
print("HTTPS=1 & Phishing=1:", Secure_Phishing)
```

```
HTTPS=0 & Phishing=0: 527664
HTTPS=0 & Phishing=1: 92235
HTTPS=1 & Phishing=0: 2325
HTTPS=1 & Phishing=1: 7621
```

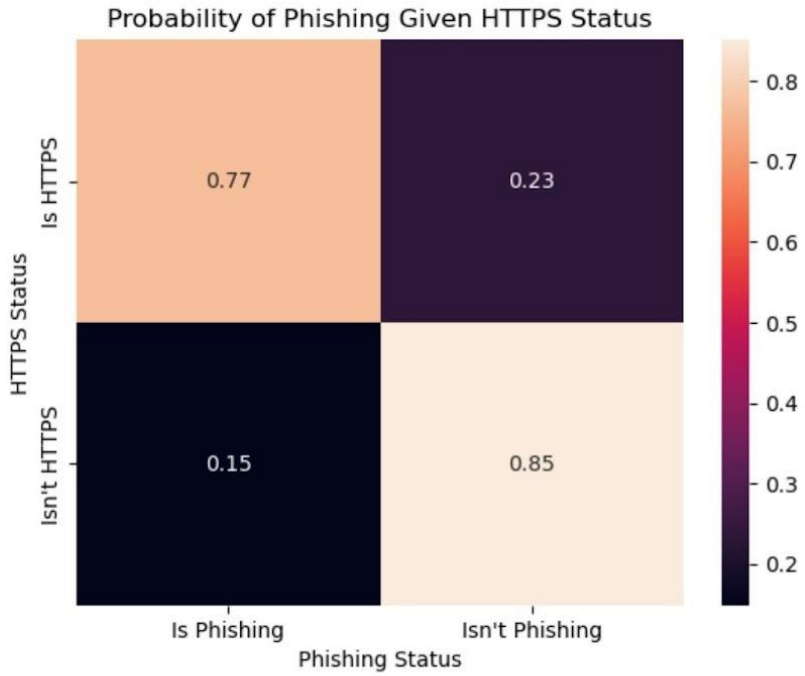
In [11]:
```
table = pd.DataFrame(
    [[Secure_Phishing, Secure_Not_Phishing],
     [Not_Secure_Phishing, Not_Secure_Not_Phishing]],
    index=["Is HTTPS", "Isn't HTTPS"],
    columns=["Is Phishing", "Isn't Phishing"])

table
```

Out[11]:

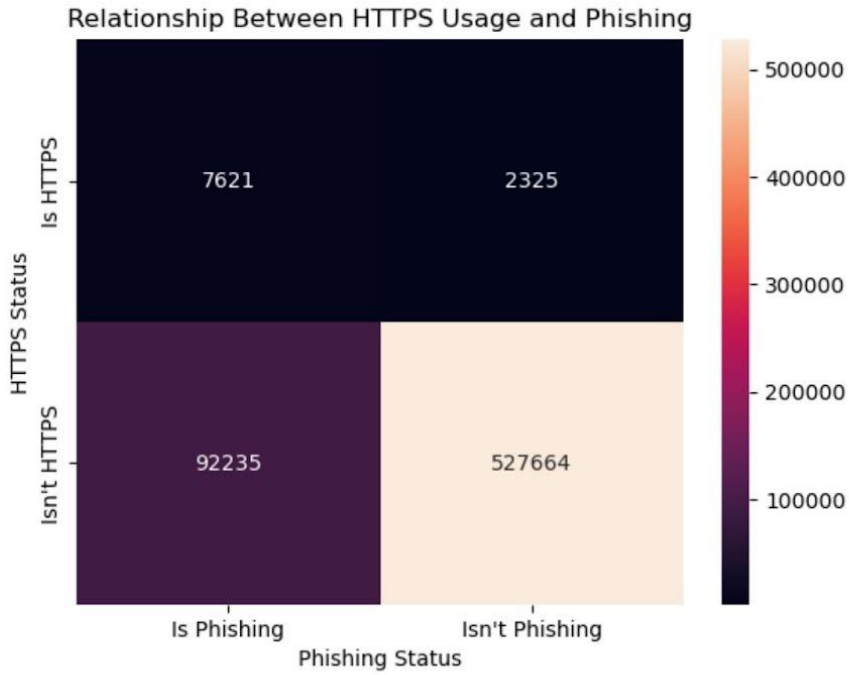
	Is Phishing	Isn't Phishing
Is HTTPS	7621	2325
Isn't HTTPS	92235	527664

In [18]:
```
plt.figure()
sns.heatmap(table, annot=True, fmt="d")
plt.title("Relationship Between HTTPS Usage and Phishing")
plt.xlabel("Phishing Status")
plt.ylabel("HTTPS Status")
plt.show()
```



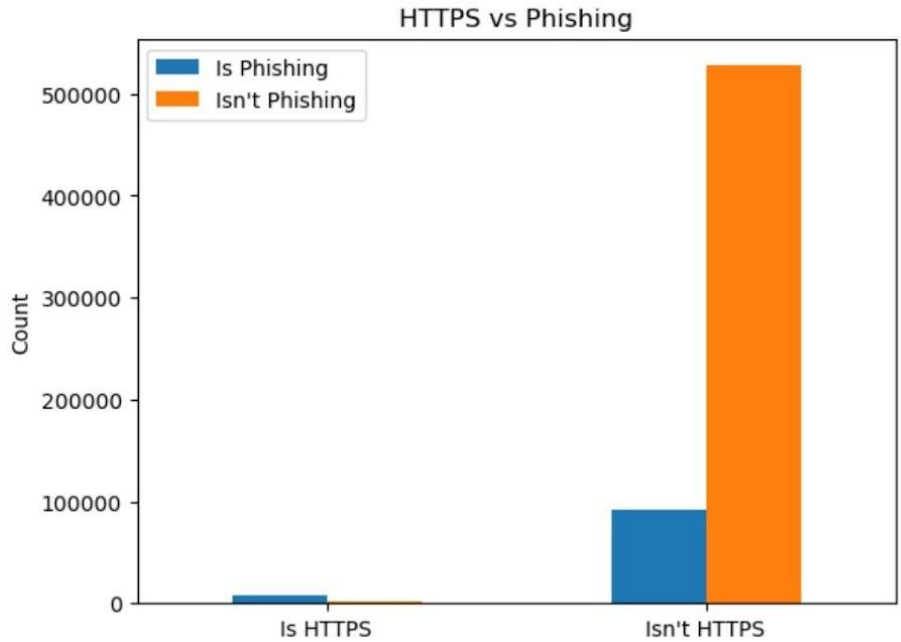
In [14]:
```
table.plot(kind="bar")

plt.title("HTTPS vs Phishing")
plt.ylabel("Count")
plt.xticks(rotation=0)
plt.show()
```



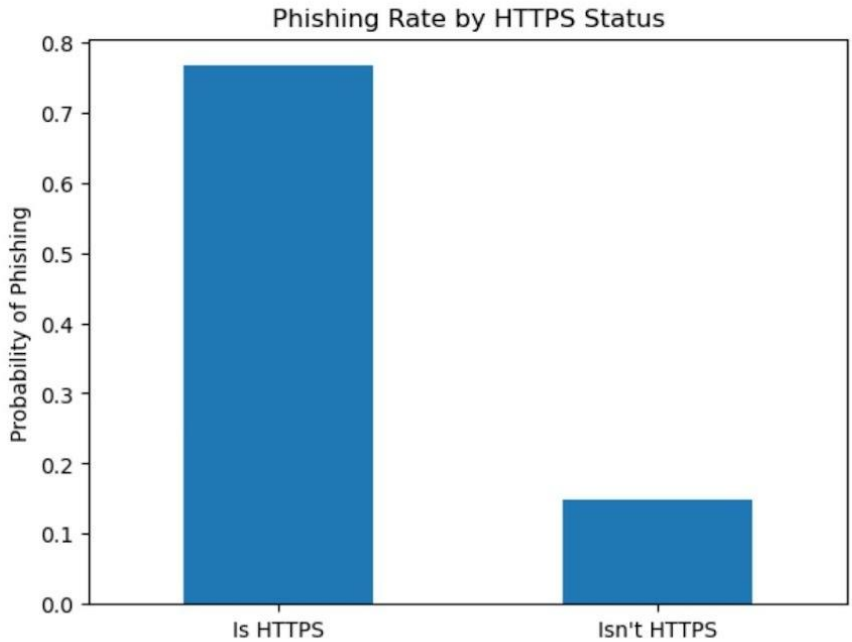
In [19]:
```
row_percent = table.div(table.sum(axis=1), axis=0)

plt.figure()
sns.heatmap(row_percent, annot=True, fmt=".2f")
plt.title("Probability of Phishing Given HTTPS Status")
plt.xlabel("Phishing Status")
plt.ylabel("HTTPS Status")
plt.show()
```



In [21]:
```
phishing_rate = row_percent["Is Phishing"]

phishing_rate.plot(kind="bar")
plt.title("Phishing Rate by HTTPS Status")
plt.ylabel("Probability of Phishing")
plt.xticks(rotation=0)
plt.show()
```



In [29]:
```
chi2, p, dof, expected = chi2_contingency(table)

print(f"Chi-square statistic: {chi2:.3f}")
print(f"Degrees of freedom: {dof}")
if p < 0.001:
    print("P-value: < 0.001")
else:
    print(f"P-value: {p:.3f}")
```

```
Chi-square statistic: 27969.842
Degrees of freedom: 1
P-value: < 0.001
```

Appendix B: #3 Comparing the relationship with URL length and phishing instances Python Code used

Import:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import matplotlib.ticker as mticker
import seaborn as sns

# Format:
sns.set_theme(style="whitegrid", font_scale=1.1)
plt.rcParams.update({
    "figure.dpi": 130,
    "axes.spines.top": False,
    "axes.spines.right": False,
    "font.family": "DejaVu Sans"
})
```

Import phishing.csv:

```
# importing phishing.csv

df = pd.read_csv("phishing.csv", index_col=0)

# Drop Rows
df = df.dropna(subset=["Phising"])
df["Phising"] = df["Phising"].astype(int)

# Columns
df["Label"] = df["Phising"].map({0: "Legitimate", 1: "Phishing"})

print(f"Dataset loaded: {len(df):,} rows × {df.shape[1]} columns")
print(f" Legitimate URLs : {(df.Phising == 0).sum():,}")
print(f" Phishing URLs   : {(df.Phising == 1).sum():,}\n")
df.head()
```

```
Dataset loaded: 630,071 rows × 13 columns
 Legitimate URLs : 530,060
 Phishing URLs   : 100,011
```

Descriptive Statistics:

```
# Step 3: Descriptive Statistics:
summary = df.groupby("Label")["UrlLength"].describe().round(2)
print("=== URL Length Summary Statistics ===")
display(summary)

legit = df[df["Phising"] == 0]["UrlLength"]
phish = df[df["Phising"] == 1]["UrlLength"]

print(f"\nMean URL Length  -->  Legitimate: {legit.mean():.1f} chars  |  Phishing: {phish.mean():.1f} chars")
print(f"Median URL Length -->  Legitimate: {legit.median():.0f} chars  |  Phishing: {phish.median():.0f} chars")
corr = df["UrlLength"].corr(df["Phising"])
print(f"Pearson Correlation (UrlLength <--> Phising): {corr:.4f}")
```

Phishing Rate by URL:

```
# Step 4: Bin URLs by length and compute phishing rate
bins   = [0, 50, 100, 200, 500, df["UrlLength"].max() + 1]
labels = ["Very Short (≤50)", "Short (51–100)", "Medium (101–200)",
          "Long (201–500)", "Very Long (500+)"]

df["LengthBin"] = pd.cut(df["UrlLength"], bins=bins, labels=labels, right=True)

bin_stats = df.groupby("LengthBin", observed=True).agg(
    Total    = ("Phising", "count"),
    Phishing = ("Phising", "sum")
)
bin_stats["Legitimate"]      = bin_stats["Total"] - bin_stats["Phishing"]
bin_stats["Phishing_Rate_%"] = (bin_stats["Phishing"] / bin_stats["Total"] * 100).round(2)

print("=== Phishing Rate by URL Length Bin ===")
display(bin_stats)
```

Histogram:

```
# Plot 1: Overlapping Histogram with Kernel Density Estimation (KDE) – shows how URL lengths are distributed for each class.
# Phishing URLs are clustering at shorter lengths
CAP = 400  # cap x-axis for readability (outliers beyond 400 are rare)

fig, ax = plt.subplots(figsize=(11, 5))

for label, color, zorder in [("Legitimate", "#2196F3", 1), ("Phishing", "#E53935", 2)]:
    data = df[df["Label"] == label]["UrlLength"]
    data_capped = data[data <= CAP]
    ax.hist(data_capped, bins=80, density=True, alpha=0.55,
            color=color, label=f"{label} (n={len(data):,})", zorder=zorder)
    # Overlay KDE
    from scipy.stats import gaussian_kde
    kde = gaussian_kde(data_capped, bw_method=0.15)
    x_range = np.linspace(0, CAP, 500)
    ax.plot(x_range, kde(x_range), color=color, linewidth=2.2, zorder=zorder+1)

# Mean lines
ax.axvline(legit.mean(), color="#1565C0", linestyle="--", linewidth=1.8, label=f"Legit mean ({legit.mean():.0f})")
ax.axvline(phish.mean(), color="#B71C1C", linestyle="--", linewidth=1.8, label=f"Phish mean ({phish.mean():.0f})")

ax.set_title("URL Length Distribution by Phishing Status", fontsize=15, fontweight="bold", pad=14)
ax.set_xlabel("URL Length (characters)", fontsize=12)
ax.set_ylabel("Density", fontsize=12)
ax.set_xlim(0, CAP)
ax.legend(fontsize=10, framealpha=0.9)
ax.grid(axis="y", alpha=0.3)

plt.tight_layout()
plt.savefig("plot1_distribution.png", dpi=150, bbox_inches="tight")
plt.show()
```

Box Plot:

```
# Plot 2: Box Plot - Median, Spread, and Outliers
fig, ax = plt.subplots(figsize=(8, 6))

df_cap = df[df["UrlLength"] <= 400].copy()
sns.boxplot(data=df_cap, x="Label", y="UrlLength",
            hue="Label", palette={"Legitimate": "#2196F3", "Phishing": "#E53935"},
            width=0.5, fliersize=2, flierprops=dict(alpha=0.15),
            legend=False, ax=ax)

# Annotate means
for i, (label, color) in enumerate([("Legitimate", "#1565C0"), ("Phishing", "#B71C1C")]):
    m = df[df["Label"] == label]["UrlLength"].mean()
    ax.text(i, m + 4, f"Mean: {m:.0f}", ha="center", fontsize=10,
            fontweight="bold", color=color)

ax.set_title("URL Length Distribution by Phishing Status\n(Box Plot, capped at 400 chars)",
             fontsize=14, fontweight="bold")
ax.set_xlabel("", fontsize=12)
ax.set_ylabel("URL Length (characters)", fontsize=12)
ax.grid(axis="y", alpha=0.3)

plt.tight_layout()
plt.savefig("plot2_boxplot.png", dpi=150, bbox_inches="tight")
plt.show()
```

Heat Map:

```
# Plot 3: Probability Heatmap Matrix
pivot = df.groupby(["LengthBin", "Label"], observed=True).size().unstack(fill_value=0)
pivot_pct = pivot.div(pivot.sum(axis=1), axis=0).round(4)

fig, ax = plt.subplots(figsize=(8, 5))
sns.heatmap(pivot_pct, annot=True, fmt=".2f", cmap="RdPu",
            linewidths=0.5, linecolor="white",
            cbar_kws={"label": "Proportion of URLs"},
            ax=ax)

ax.set_title("Probability Matrix: URL Length Bin vs. Phishing Status",
             fontsize=13, fontweight="bold", pad=14)
ax.set_xlabel("Phishing Status", fontsize=11)
ax.set_ylabel("URL Length Bin", fontsize=11)
ax.set_xticklabels(["Legitimate", "Phishing"], fontsize=10)
ax.set_yticklabels(ax.get_yticklabels(), rotation=0, fontsize=10)

plt.tight_layout()
plt.savefig("plot3_heatmap.png", dpi=150, bbox_inches="tight")
plt.show()
```

Bar Chart:

```
# Plot 4: Phishing Rate Bar Chart - Phishing Rate by URL Length Bin - (Which URL Length ranges are most dangerous?)
rates  = bin_stats["Phishing_Rate_%"]
colors = ["#4CAF50" if r < 15 else "#FF9800" if r < 22 else "#E53935" for r in rates]

fig, ax = plt.subplots(figsize=(10, 5))
bars = ax.bar(bin_stats.index, rates, color=colors, edgecolor="white",
              linewidth=0.8, width=0.55)

# Value labels
for bar, val in zip(bars, rates):
    ax.text(bar.get_x() + bar.get_width() / 2,
            bar.get_height() + 0.3,
            f"{val:.1f}%", ha="center", va="bottom",
            fontsize=11, fontweight="bold")

ax.set_title("Phishing Rate (%) by URL Length Bin", fontsize=14, fontweight="bold")
ax.set_xlabel("URL Length Bin", fontsize=11)
ax.set_ylabel("Phishing Rate (%)", fontsize=11)
ax.set_ylim(0, rates.max() * 1.25)
ax.grid(axis="y", alpha=0.3, linestyle="--")

patches = [
    mpatches.Patch(color="#4CAF50", label="Low risk (<15%)"),
    mpatches.Patch(color="#FF9800", label="Medium risk (15–22%)"),
    mpatches.Patch(color="#E53935", label="High risk (>22%)"),
]
ax.legend(handles=patches, fontsize=9, loc="upper right")

plt.tight_layout()
plt.savefig("plot4_phishing_rate.png", dpi=150, bbox_inches="tight")
plt.show()
```

Stacked Bar Chart:

```
# Plot 5: Stacked Bar Chart — Volume Composition per Bin
fig, ax = plt.subplots(figsize=(10, 5))

bin_counts = df.groupby(["LengthBin", "Label"], observed=True).size().unstack(fill_value=0)
bin_counts[["Legitimate", "Phishing"]].plot(
    kind="bar", stacked=True, ax=ax,
    color=["#2196F3", "#E53935"], edgecolor="white", linewidth=0.5, width=0.6
)

ax.set_title("URL Count by Length Bin — Legitimate vs. Phishing (Stacked)",
             fontsize=13, fontweight="bold")
ax.set_xlabel("URL Length Bin", fontsize=11)
ax.set_ylabel("Number of URLs", fontsize=11)
ax.set_xticklabels(ax.get_xticklabels(), rotation=20, ha="right")
ax.yaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _: f"{int(x):,}"))
ax.legend(title="Class", fontsize=10)
ax.grid(axis="y", alpha=0.3, linestyle="--")

plt.tight_layout()
plt.savefig("plot5_stacked_bar.png", dpi=150, bbox_inches="tight")
plt.show()
```

Interpretation:

```
# Key Findings & Interpretation:

top_bin  = bin_stats["Phishing_Rate_%"].idxmax()
top_rate = bin_stats["Phishing_Rate_%"].max()

print("=" * 60)
print("         URL LENGTH vs. PHISHING — KEY FINDINGS")
print("=" * 60)
print(f"  Total URLs analyzed     : {len(df):,}")
print(f"  Legitimate              : {(df.Phising==0).sum():,} ({(df.Phising==0).mean()*100:.1f}%)")
print(f"  Phishing                : {(df.Phising==1).sum():,} ({(df.Phising==1).mean()*100:.1f}%)")
print()
print(f"  Avg length — Legitimate : {legit.mean():.1f} chars")
print(f"  Avg length — Phishing   : {phish.mean():.1f} chars")
print(f"  Difference              : {legit.mean()-phish.mean():.1f} chars (legit URLs are LONGER)")
print()
print(f"  Pearson correlation     : {df['UrlLength'].corr(df['Phising']):.4f} (negative = longer -> less phishing)")
print(f"  Highest-risk bin        : {top_bin} -> {top_rate:.1f}% phishing rate")
print()
print("  INTERPRETATION:")
print("  Contrary to common assumptions, phishing URLs in this")
print("  dataset tend to be SHORTER than legitimate ones. Very")
print("  short URLs (≤50 chars) have a ~22% phishing rate, and")
print("  very long URLs (500+) spike again at ~26%. The safest")
print("  range is 51–200 characters (~8–9% phishing rate).")
print("=" * 60)
```

AI Disclosure

The ChatGPT LLM was used during this project. The uses included aid in programing the python to create the data visualizations. It was also used to find relevant sources that would corroborate my findings. The LLM was also used to review my work for errors and consistency in tone and content.

It gave me general ideas on what are the best graphs to visualize data from the selected phishing dataset and explained why these graphs were the best method. To help make data be seen in the best way with explanations. It told me that a scatterplot and histogram were the best ways to display my data given the variables that I wanted to demonstrate. When I created these graphs, it worked smoothly with what I wanted to display. No AI-generated text was copied verbatim in my assignment.

Additionally, used for additional interpretation of the observations' assistance and suggestion of the best fit of the graph for the visualization analysis.

Finally, prompt used for generation of grid visual for the Mean, Median and Standard Deviation and how to best structure the description of the charts and what they show.

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